



# GT\_Plate6\_Multiplex\_12\_18\_2024.pcrd

3/27/2025 9:48 AM

## Report Information

User: BioRad/admin  
Data File Name: GT\_Plate6\_Multiplex\_12\_18\_2024.pcrd  
Data File Path: C:\Users\CSMTSS1\Documents\Tortoises\Plates\Multi\_SCNAG\_Mt  
Well Group Name: All Wells  
Report Differs from Last Save: No

## Run Setup

### Run Information

Run Date: 12/18/2024 12:24 PM  
Run User: admin  
Run Type: User-defined  
Plate File: Elgaria\_Multiplex.prcl.pltd  
ID:  
Notes:  
Sample Volume: 25  
Temperature Control Mode: Calculated  
Lid Temperature: 105  
Base Serial Number: 795BR20744  
Optical Head Serial Number: 795BR20744

### Protocol

- 1: 95.0°C for 3:00
- 2: 95.0°C for 0:15
- 3: 62.0°C for 0:30  
Plate Read
- 4: GOTO 2, 44 more times

### Plate Display

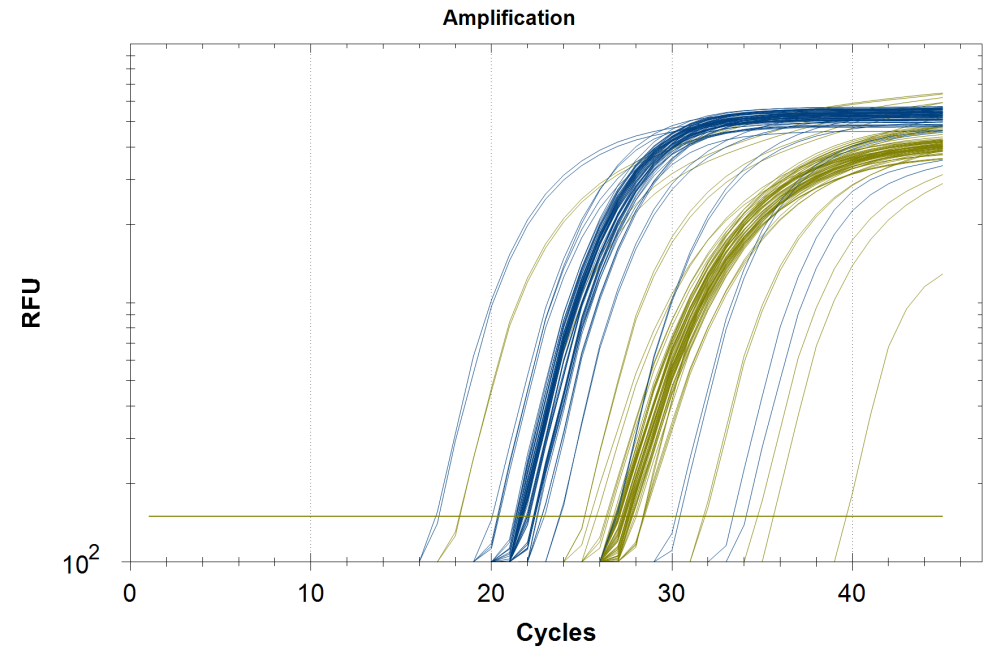
|   | 1                           | 2                           | 3                              | 4                              | 5                              | 6                              | 7                              | 8                              | 9                              | 10                             | 11                             | 12                             |
|---|-----------------------------|-----------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| A | Std-1<br>FAM<br>VIC<br>STD1 | Std-1<br>FAM<br>VIC<br>STD1 | *Unk-1<br>FAM<br>VIC<br>6.1_B  | *Unk-1<br>FAM<br>VIC<br>6.1_B  | *Unk-9<br>FAM<br>VIC<br>14.4_D | *Unk-9<br>FAM<br>VIC<br>14.4_D | Unk-17<br>FAM<br>VIC<br>9.2_B  | Unk-17<br>FAM<br>VIC<br>9.2_B  | Unk-25<br>FAM<br>VIC<br>10.3_C | Unk-25<br>FAM<br>VIC<br>10.3_C | Unk-33<br>FAM<br>VIC<br>11.4_B | Unk-33<br>FAM<br>VIC<br>11.4_B |
| B | Std-2<br>FAM<br>VIC<br>STD2 | Std-2<br>FAM<br>VIC<br>STD2 | Unk-2<br>FAM<br>VIC<br>15.1_A  | Unk-2<br>FAM<br>VIC<br>15.1_A  | Unk-10<br>FAM<br>VIC<br>10.5_C | Unk-10<br>FAM<br>VIC<br>10.5_C | Unk-18<br>FAM<br>VIC<br>7.5_B  | Unk-18<br>FAM<br>VIC<br>7.5_B  | Unk-26<br>FAM<br>VIC<br>11.3_C | Unk-26<br>FAM<br>VIC<br>11.3_C | Unk-34<br>FAM<br>VIC<br>10.4_A | Unk-34<br>FAM<br>VIC<br>10.4_A |
| C | Std-3<br>FAM<br>VIC<br>STD3 | Std-3<br>FAM<br>VIC<br>STD3 | *Unk-3<br>FAM<br>VIC<br>15.4_C | *Unk-3<br>FAM<br>VIC<br>15.4_C | Unk-11<br>FAM<br>VIC<br>15.5_C | Unk-11<br>FAM<br>VIC<br>15.5_C | Unk-19<br>FAM<br>VIC<br>6.5_B  | Unk-19<br>FAM<br>VIC<br>6.5_B  | *Unk-27<br>FAM<br>VIC<br>6.1_A | *Unk-27<br>FAM<br>VIC<br>6.1_A | Unk-35<br>FAM<br>VIC<br>7.2_A  | Unk-35<br>FAM<br>VIC<br>7.2_A  |
| D | Std-4<br>FAM<br>VIC<br>STD4 | Std-4<br>FAM<br>VIC<br>STD4 | Unk-4<br>FAM<br>VIC<br>9.3_D   | Unk-4<br>FAM<br>VIC<br>9.3_D   | Unk-12<br>FAM<br>VIC<br>10.2_C | Unk-12<br>FAM<br>VIC<br>10.2_C | *Unk-20<br>FAM<br>VIC<br>6.1_C | *Unk-20<br>FAM<br>VIC<br>6.1_C | *Unk-28<br>FAM<br>VIC<br>6.1_D | *Unk-28<br>FAM<br>VIC<br>6.1_D | Unk-36<br>FAM<br>VIC<br>5.6_B  | Unk-36<br>FAM<br>VIC<br>5.6_B  |

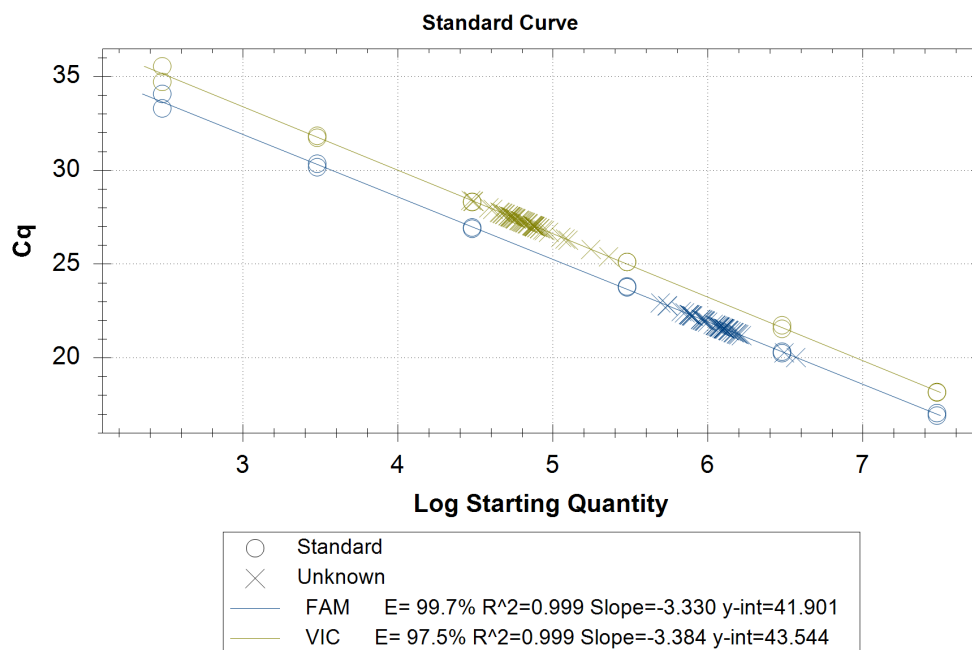
Plate Display

|   | 1                           | 2                           | 3                             | 4                             | 5                             | 6                             | 7                              | 8                              | 9                              | 10                             | 11                             | 12                             |
|---|-----------------------------|-----------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| E | Std-5<br>FAM<br>VIC<br>STD5 | Std-5<br>FAM<br>VIC<br>STD5 | Unk-5<br>FAM<br>VIC<br>12.5_D | Unk-5<br>FAM<br>VIC<br>12.5_D | Unk-13<br>FAM<br>VIC<br>6.5_C | Unk-13<br>FAM<br>VIC<br>6.5_C | Unk-21<br>FAM<br>VIC<br>3.4_D  | Unk-21<br>FAM<br>VIC<br>3.4_D  | Unk-29<br>FAM<br>VIC<br>10.5_A | Unk-29<br>FAM<br>VIC<br>10.5_A | Unk-37<br>FAM<br>VIC<br>9.2_D  | Unk-37<br>FAM<br>VIC<br>9.2_D  |
| F | Std-6<br>FAM<br>VIC<br>STD6 | Std-6<br>FAM<br>VIC<br>STD6 | *Unk-6<br>FAM<br>VIC<br>5.2_A | *Unk-6<br>FAM<br>VIC<br>5.2_A | Unk-14<br>FAM<br>VIC<br>5.4_A | Unk-14<br>FAM<br>VIC<br>5.4_A | Unk-22<br>FAM<br>VIC<br>13.1_B | Unk-22<br>FAM<br>VIC<br>13.1_B | Unk-30<br>FAM<br>VIC<br>6.2_D  | Unk-30<br>FAM<br>VIC<br>6.2_D  | Unk-38<br>FAM<br>VIC<br>13.2_A | Unk-38<br>FAM<br>VIC<br>13.2_A |
| G |                             |                             | *Unk-7<br>FAM<br>VIC<br>6.3_A | *Unk-7<br>FAM<br>VIC<br>6.3_A | Unk-15<br>FAM<br>VIC<br>2.3_B | Unk-15<br>FAM<br>VIC<br>2.3_B | Unk-23<br>FAM<br>VIC<br>12.4_C | Unk-23<br>FAM<br>VIC<br>12.4_C | Unk-31<br>FAM<br>VIC<br>5.2_D  | Unk-31<br>FAM<br>VIC<br>5.2_D  | *Unk-39<br>FAM<br>VIC<br>5.6_A | *Unk-39<br>FAM<br>VIC<br>5.6_A |
| H | NTC-1<br>FAM<br>VIC<br>NEG  | NTC-1<br>FAM<br>VIC<br>NEG  | Unk-8<br>FAM<br>VIC<br>9.3_B  | Unk-8<br>FAM<br>VIC<br>9.3_B  | Unk-16<br>FAM<br>VIC<br>7.4_B | Unk-16<br>FAM<br>VIC<br>7.4_B | *Unk-24<br>FAM<br>VIC<br>5.6_D | *Unk-24<br>FAM<br>VIC<br>5.6_D | Unk-32<br>FAM<br>VIC<br>6.3_C  | Unk-32<br>FAM<br>VIC<br>6.3_C  | Unk-40<br>FAM<br>VIC<br>15.5_D | Unk-40<br>FAM<br>VIC<br>15.5_D |

Quantification

Step #: 3  
Analysis Mode: Fluorophore  
Baseline Setting: Baseline Subtracted Curve Fit  
Cq Determination: Single Threshold  
Baseline Method:  
FAM: Auto Calculated  
VIC: Auto Calculated  
Threshold Setting:  
FAM: 150.00, User Defined  
VIC: 150.00, User Defined





## Quantification Data

| Well | Fluor | Target | Content | Sample | Cq    | Cq Mean | Cq Std. Dev | Starting Quantity (SQ) | Log Starting Quantity | SQ Mean  | SQ Std. Dev |
|------|-------|--------|---------|--------|-------|---------|-------------|------------------------|-----------------------|----------|-------------|
| A01  | FAM   |        | Std-01  | STD1   | 17.06 | 17.00   | 0.093       | 3.000E+07              | 7.477                 | 3.00E+07 | 0.00E+00    |
| A02  | FAM   |        | Std-01  | STD1   | 16.93 | 17.00   | 0.093       | 3.000E+07              | 7.477                 | 3.00E+07 | 0.00E+00    |
| A07  | FAM   |        | Unkn-17 | 9.2_B  | 21.57 | 21.58   | 0.013       | 1.274E+06              | 6.105                 | 1.27E+06 | 1.13E+04    |
| A08  | FAM   |        | Unkn-17 | 9.2_B  | 21.59 | 21.58   | 0.013       | 1.258E+06              | 6.100                 | 1.27E+06 | 1.13E+04    |
| A09  | FAM   |        | Unkn-25 | 10.3_C | 22.42 | 22.46   | 0.051       | 7.083E+05              | 5.850                 | 6.91E+05 | 2.43E+04    |
| A10  | FAM   |        | Unkn-25 | 10.3_C | 22.49 | 22.46   | 0.051       | 6.739E+05              | 5.829                 | 6.91E+05 | 2.43E+04    |
| A11  | FAM   |        | Unkn-33 | 11.4_B | 21.99 | 22.00   | 0.013       | 9.575E+05              | 5.981                 | 9.52E+05 | 8.34E+03    |
| A12  | FAM   |        | Unkn-33 | 11.4_B | 22.00 | 22.00   | 0.013       | 9.457E+05              | 5.976                 | 9.52E+05 | 8.34E+03    |
| B01  | FAM   |        | Std-02  | STD2   | 20.34 | 20.30   | 0.056       | 3.000E+06              | 6.477                 | 3.00E+06 | 0.00E+00    |
| B02  | FAM   |        | Std-02  | STD2   | 20.26 | 20.30   | 0.056       | 3.000E+06              | 6.477                 | 3.00E+06 | 0.00E+00    |
| B03  | FAM   |        | Unkn-02 | 15.1_A | 20.03 | 20.16   | 0.179       | 3.693E+06              | 6.567                 | 3.40E+06 | 4.19E+05    |
| B04  | FAM   |        | Unkn-02 | 15.1_A | 20.29 | 20.16   | 0.179       | 3.101E+06              | 6.491                 | 3.40E+06 | 4.19E+05    |
| B05  | FAM   |        | Unkn-10 | 10.5_C | 21.47 | 21.47   | 0.008       | 1.372E+06              | 6.137                 | 1.37E+06 | 7.61E+03    |
| B06  | FAM   |        | Unkn-10 | 10.5_C | 21.48 | 21.47   | 0.008       | 1.362E+06              | 6.134                 | 1.37E+06 | 7.61E+03    |
| B07  | FAM   |        | Unkn-18 | 7.5_B  | 21.36 | 21.33   | 0.036       | 1.478E+06              | 6.170                 | 1.50E+06 | 3.72E+04    |
| B08  | FAM   |        | Unkn-18 | 7.5_B  | 21.31 | 21.33   | 0.036       | 1.531E+06              | 6.185                 | 1.50E+06 | 3.72E+04    |
| B09  | FAM   |        | Unkn-26 | 11.3_C | 21.68 | 21.75   | 0.089       | 1.182E+06              | 6.072                 | 1.13E+06 | 6.98E+04    |
| B10  | FAM   |        | Unkn-26 | 11.3_C | 21.81 | 21.75   | 0.089       | 1.083E+06              | 6.035                 | 1.13E+06 | 6.98E+04    |
| B11  | FAM   |        | Unkn-34 | 10.4_A | 21.70 | 21.68   | 0.025       | 1.166E+06              | 6.067                 | 1.18E+06 | 2.02E+04    |
| B12  | FAM   |        | Unkn-34 | 10.4_A | 21.67 | 21.68   | 0.025       | 1.195E+06              | 6.077                 | 1.18E+06 | 2.02E+04    |
| C01  | FAM   |        | Std-03  | STD3   | 23.77 | 23.79   | 0.035       | 3.000E+05              | 5.477                 | 3.00E+05 | 0.00E+00    |
| C02  | FAM   |        | Std-03  | STD3   | 23.82 | 23.79   | 0.035       | 3.000E+05              | 5.477                 | 3.00E+05 | 0.00E+00    |
| C05  | FAM   |        | Unkn-11 | 15.5_C | 22.33 | 22.32   | 0.015       | 7.551E+05              | 5.878                 | 7.61E+05 | 8.01E+03    |
| C06  | FAM   |        | Unkn-11 | 15.5_C | 22.31 | 22.32   | 0.015       | 7.664E+05              | 5.884                 | 7.61E+05 | 8.01E+03    |
| C07  | FAM   |        | Unkn-19 | 6.5_B  | 21.58 | 21.57   | 0.005       | 1.271E+06              | 6.104                 | 1.27E+06 | 4.41E+03    |
| C08  | FAM   |        | Unkn-19 | 6.5_B  | 21.57 | 21.57   | 0.005       | 1.277E+06              | 6.106                 | 1.27E+06 | 4.41E+03    |
| C11  | FAM   |        | Unkn-35 | 7.2_A  | 21.26 | 21.42   | 0.235       | 1.587E+06              | 6.200                 | 1.42E+06 | 2.30E+05    |
| C12  | FAM   |        | Unkn-35 | 7.2_A  | 21.59 | 21.42   | 0.235       | 1.261E+06              | 6.101                 | 1.42E+06 | 2.30E+05    |

## Quantification Data

| Well | Fluor | Target | Content | Sample | Cq    | Cq Mean | Cq Std. Dev | Starting Quantity (SQ) | Log Starting Quantity | SQ Mean  | SQ Std. Dev |
|------|-------|--------|---------|--------|-------|---------|-------------|------------------------|-----------------------|----------|-------------|
| D01  | FAM   |        | Std-04  | STD4   | 26.96 | 26.93   | 0.050       | 3.000E+04              | 4.477                 | 3.00E+04 | 0.00E+00    |
| D02  | FAM   |        | Std-04  | STD4   | 26.89 | 26.93   | 0.050       | 3.000E+04              | 4.477                 | 3.00E+04 | 0.00E+00    |
| D03  | FAM   |        | Unkn-04 | 9.3_D  | 22.03 | 22.00   | 0.038       | 9.286E+05              | 5.968                 | 9.46E+05 | 2.45E+04    |
| D04  | FAM   |        | Unkn-04 | 9.3_D  | 21.98 | 22.00   | 0.038       | 9.633E+05              | 5.984                 | 9.46E+05 | 2.45E+04    |
| D05  | FAM   |        | Unkn-12 | 10.2_C | 21.55 | 21.57   | 0.028       | 1.292E+06              | 6.111                 | 1.27E+06 | 2.45E+04    |
| D06  | FAM   |        | Unkn-12 | 10.2_C | 21.59 | 21.57   | 0.028       | 1.257E+06              | 6.099                 | 1.27E+06 | 2.45E+04    |
| D11  | FAM   |        | Unkn-36 | 5.6_B  | 21.67 | 21.67   | 0.002       | 1.193E+06              | 6.077                 | 1.19E+06 | 1.28E+03    |
| D12  | FAM   |        | Unkn-36 | 5.6_B  | 21.67 | 21.67   | 0.002       | 1.192E+06              | 6.076                 | 1.19E+06 | 1.28E+03    |
| E01  | FAM   |        | Std-05  | STD5   | 30.36 | 30.27   | 0.134       | 3.000E+03              | 3.477                 | 3.00E+03 | 0.00E+00    |
| E02  | FAM   |        | Std-05  | STD5   | 30.17 | 30.27   | 0.134       | 3.000E+03              | 3.477                 | 3.00E+03 | 0.00E+00    |
| E03  | FAM   |        | Unkn-05 | 12.5_D | 21.58 | 21.56   | 0.018       | 1.272E+06              | 6.105                 | 1.28E+06 | 1.60E+04    |
| E04  | FAM   |        | Unkn-05 | 12.5_D | 21.55 | 21.56   | 0.018       | 1.295E+06              | 6.112                 | 1.28E+06 | 1.60E+04    |
| E05  | FAM   |        | Unkn-13 | 6.5_C  | 22.31 | 22.28   | 0.045       | 7.650E+05              | 5.884                 | 7.82E+05 | 2.41E+04    |
| E06  | FAM   |        | Unkn-13 | 6.5_C  | 22.25 | 22.28   | 0.045       | 7.991E+05              | 5.903                 | 7.82E+05 | 2.41E+04    |
| E07  | FAM   |        | Unkn-21 | 3.4_D  | 21.37 | 21.38   | 0.002       | 1.462E+06              | 6.165                 | 1.46E+06 | 1.91E+03    |
| E08  | FAM   |        | Unkn-21 | 3.4_D  | 21.38 | 21.38   | 0.002       | 1.459E+06              | 6.164                 | 1.46E+06 | 1.91E+03    |
| E09  | FAM   |        | Unkn-29 | 10.5_A | 22.08 | 21.92   | 0.230       | 8.973E+05              | 5.953                 | 1.01E+06 | 1.60E+05    |
| E10  | FAM   |        | Unkn-29 | 10.5_A | 21.75 | 21.92   | 0.230       | 1.124E+06              | 6.051                 | 1.01E+06 | 1.60E+05    |
| E11  | FAM   |        | Unkn-37 | 9.2_D  | 22.27 | 22.27   | 0.004       | 7.847E+05              | 5.895                 | 7.86E+05 | 1.99E+03    |
| E12  | FAM   |        | Unkn-37 | 9.2_D  | 22.27 | 22.27   | 0.004       | 7.875E+05              | 5.896                 | 7.86E+05 | 1.99E+03    |
| F01  | FAM   |        | Std-06  | STD6   | 33.31 | 33.69   | 0.546       | 3.000E+02              | 2.477                 | 3.00E+02 | 0.00E+00    |
| F02  | FAM   |        | Std-06  | STD6   | 34.08 | 33.69   | 0.546       | 3.000E+02              | 2.477                 | 3.00E+02 | 0.00E+00    |
| F05  | FAM   |        | Unkn-14 | 5.4_A  | 21.66 | 22.00   | 0.481       | 1.204E+06              | 6.080                 | 9.78E+05 | 3.19E+05    |
| F06  | FAM   |        | Unkn-14 | 5.4_A  | 22.34 | 22.00   | 0.481       | 7.520E+05              | 5.876                 | 9.78E+05 | 3.19E+05    |
| F07  | FAM   |        | Unkn-22 | 13.1_B | 21.58 | 21.54   | 0.061       | 1.269E+06              | 6.103                 | 1.31E+06 | 5.54E+04    |
| F08  | FAM   |        | Unkn-22 | 13.1_B | 21.49 | 21.54   | 0.061       | 1.347E+06              | 6.129                 | 1.31E+06 | 5.54E+04    |
| F09  | FAM   |        | Unkn-30 | 6.2_D  | 21.28 | 21.29   | 0.018       | 1.563E+06              | 6.194                 | 1.55E+06 | 1.93E+04    |
| F10  | FAM   |        | Unkn-30 | 6.2_D  | 21.30 | 21.29   | 0.018       | 1.536E+06              | 6.186                 | 1.55E+06 | 1.93E+04    |
| F11  | FAM   |        | Unkn-38 | 13.2_A | 22.07 | 22.43   | 0.513       | 9.025E+05              | 5.955                 | 7.25E+05 | 2.52E+05    |
| F12  | FAM   |        | Unkn-38 | 13.2_A | 22.80 | 22.43   | 0.513       | 5.465E+05              | 5.738                 | 7.25E+05 | 2.52E+05    |
| G05  | FAM   |        | Unkn-15 | 2.3_B  | 21.93 | 21.83   | 0.141       | 9.980E+05              | 5.999                 | 1.07E+06 | 1.05E+05    |
| G06  | FAM   |        | Unkn-15 | 2.3_B  | 21.73 | 21.83   | 0.141       | 1.146E+06              | 6.059                 | 1.07E+06 | 1.05E+05    |
| G07  | FAM   |        | Unkn-23 | 12.4_C | 21.20 | 21.48   | 0.397       | 1.647E+06              | 6.217                 | 1.38E+06 | 3.75E+05    |
| G08  | FAM   |        | Unkn-23 | 12.4_C | 21.76 | 21.48   | 0.397       | 1.118E+06              | 6.048                 | 1.38E+06 | 3.75E+05    |
| G09  | FAM   |        | Unkn-31 | 5.2_D  | 21.49 | 21.46   | 0.045       | 1.352E+06              | 6.131                 | 1.38E+06 | 4.32E+04    |
| G10  | FAM   |        | Unkn-31 | 5.2_D  | 21.42 | 21.46   | 0.045       | 1.413E+06              | 6.150                 | 1.38E+06 | 4.32E+04    |
| H01  | FAM   |        | NTC-01  | NEG    | N/A   | 0.00    | 0.000       | N/A                    | N/A                   | 0.00E+00 | 0.00E+00    |
| H02  | FAM   |        | NTC-01  | NEG    | N/A   | 0.00    | 0.000       | N/A                    | N/A                   | 0.00E+00 | 0.00E+00    |
| H03  | FAM   |        | Unkn-08 | 9.3_B  | 22.33 | 22.30   | 0.047       | 7.547E+05              | 5.878                 | 7.72E+05 | 2.51E+04    |
| H04  | FAM   |        | Unkn-08 | 9.3_B  | 22.26 | 22.30   | 0.047       | 7.902E+05              | 5.898                 | 7.72E+05 | 2.51E+04    |
| H05  | FAM   |        | Unkn-16 | 7.4_B  | 21.62 | 21.69   | 0.094       | 1.232E+06              | 6.091                 | 1.18E+06 | 7.66E+04    |
| H06  | FAM   |        | Unkn-16 | 7.4_B  | 21.75 | 21.69   | 0.094       | 1.124E+06              | 6.051                 | 1.18E+06 | 7.66E+04    |
| H09  | FAM   |        | Unkn-32 | 6.3_C  | 22.04 | 22.01   | 0.048       | 9.200E+05              | 5.964                 | 9.42E+05 | 3.10E+04    |
| H10  | FAM   |        | Unkn-32 | 6.3_C  | 21.98 | 22.01   | 0.048       | 9.639E+05              | 5.984                 | 9.42E+05 | 3.10E+04    |
| H11  | FAM   |        | Unkn-40 | 15.5_D | 22.79 | 22.87   | 0.112       | 5.501E+05              | 5.740                 | 5.22E+05 | 4.03E+04    |
| H12  | FAM   |        | Unkn-40 | 15.5_D | 22.95 | 22.87   | 0.112       | 4.931E+05              | 5.693                 | 5.22E+05 | 4.03E+04    |
| A01  | VIC   |        | Std-01  | STD1   | 18.19 | 18.18   | 0.021       | 3.000E+07              | 7.477                 | 3.00E+07 | 0.00E+00    |
| A02  | VIC   |        | Std-01  | STD1   | 18.16 | 18.18   | 0.021       | 3.000E+07              | 7.477                 | 3.00E+07 | 0.00E+00    |
| A07  | VIC   |        | Unkn-17 | 9.2_B  | 27.18 | 27.22   | 0.055       | 6.863E+04              | 4.837                 | 6.69E+04 | 2.50E+03    |

## Quantification Data

| Well | Fluor | Target | Content | Sample | Cq    | Cq Mean | Cq Std. Dev | Starting Quantity (SQ) | Log Starting Quantity | SQ Mean  | SQ Std. Dev |
|------|-------|--------|---------|--------|-------|---------|-------------|------------------------|-----------------------|----------|-------------|
| A08  | VIC   |        | Unkn-17 | 9.2_B  | 27.25 | 27.22   | 0.055       | 6.509E+04              | 4.814                 | 6.69E+04 | 2.50E+03    |
| A09  | VIC   |        | Unkn-25 | 10.3_C | 27.51 | 27.55   | 0.057       | 5.456E+04              | 4.737                 | 5.31E+04 | 2.06E+03    |
| A10  | VIC   |        | Unkn-25 | 10.3_C | 27.59 | 27.55   | 0.057       | 5.165E+04              | 4.713                 | 5.31E+04 | 2.06E+03    |
| A11  | VIC   |        | Unkn-33 | 11.4_B | 27.31 | 27.33   | 0.032       | 6.264E+04              | 4.797                 | 6.17E+04 | 1.36E+03    |
| A12  | VIC   |        | Unkn-33 | 11.4_B | 27.36 | 27.33   | 0.032       | 6.072E+04              | 4.783                 | 6.17E+04 | 1.36E+03    |
| B01  | VIC   |        | Std-02  | STD2   | 21.75 | 21.65   | 0.136       | 3.000E+06              | 6.477                 | 3.00E+06 | 0.00E+00    |
| B02  | VIC   |        | Std-02  | STD2   | 21.56 | 21.65   | 0.136       | 3.000E+06              | 6.477                 | 3.00E+06 | 0.00E+00    |
| B03  | VIC   |        | Unkn-02 | 15.1_A | 25.41 | 25.60   | 0.275       | 2.285E+05              | 5.359                 | 2.02E+05 | 3.76E+04    |
| B04  | VIC   |        | Unkn-02 | 15.1_A | 25.80 | 25.60   | 0.275       | 1.754E+05              | 5.244                 | 2.02E+05 | 3.76E+04    |
| B05  | VIC   |        | Unkn-10 | 10.5_C | 26.72 | 26.76   | 0.064       | 9.384E+04              | 4.972                 | 9.10E+04 | 3.99E+03    |
| B06  | VIC   |        | Unkn-10 | 10.5_C | 26.81 | 26.76   | 0.064       | 8.820E+04              | 4.945                 | 9.10E+04 | 3.99E+03    |
| B07  | VIC   |        | Unkn-18 | 7.5_B  | 26.92 | 26.88   | 0.050       | 8.200E+04              | 4.914                 | 8.40E+04 | 2.83E+03    |
| B08  | VIC   |        | Unkn-18 | 7.5_B  | 26.84 | 26.88   | 0.050       | 8.601E+04              | 4.935                 | 8.40E+04 | 2.83E+03    |
| B09  | VIC   |        | Unkn-26 | 11.3_C | 26.91 | 26.94   | 0.044       | 8.238E+04              | 4.916                 | 8.07E+04 | 2.44E+03    |
| B10  | VIC   |        | Unkn-26 | 11.3_C | 26.97 | 26.94   | 0.044       | 7.893E+04              | 4.897                 | 8.07E+04 | 2.44E+03    |
| B11  | VIC   |        | Unkn-34 | 10.4_A | 27.06 | 27.04   | 0.033       | 7.414E+04              | 4.870                 | 7.53E+04 | 1.69E+03    |
| B12  | VIC   |        | Unkn-34 | 10.4_A | 27.02 | 27.04   | 0.033       | 7.653E+04              | 4.884                 | 7.53E+04 | 1.69E+03    |
| C01  | VIC   |        | Std-03  | STD3   | 25.12 | 25.12   | 0.000       | 3.000E+05              | 5.477                 | 3.00E+05 | 0.00E+00    |
| C02  | VIC   |        | Std-03  | STD3   | 25.12 | 25.12   | 0.000       | 3.000E+05              | 5.477                 | 3.00E+05 | 0.00E+00    |
| C05  | VIC   |        | Unkn-11 | 15.5_C | 27.94 | 27.86   | 0.114       | 4.094E+04              | 4.612                 | 4.33E+04 | 3.36E+03    |
| C06  | VIC   |        | Unkn-11 | 15.5_C | 27.77 | 27.86   | 0.114       | 4.570E+04              | 4.660                 | 4.33E+04 | 3.36E+03    |
| C07  | VIC   |        | Unkn-19 | 6.5_B  | 27.03 | 27.00   | 0.046       | 7.570E+04              | 4.879                 | 7.74E+04 | 2.41E+03    |
| C08  | VIC   |        | Unkn-19 | 6.5_B  | 26.97 | 27.00   | 0.046       | 7.911E+04              | 4.898                 | 7.74E+04 | 2.41E+03    |
| C11  | VIC   |        | Unkn-35 | 7.2_A  | 27.23 | 27.37   | 0.190       | 6.598E+04              | 4.819                 | 6.05E+04 | 7.81E+03    |
| C12  | VIC   |        | Unkn-35 | 7.2_A  | 27.50 | 27.37   | 0.190       | 5.494E+04              | 4.740                 | 6.05E+04 | 7.81E+03    |
| D01  | VIC   |        | Std-04  | STD4   | 28.33 | 28.32   | 0.010       | 3.000E+04              | 4.477                 | 3.00E+04 | 0.00E+00    |
| D02  | VIC   |        | Std-04  | STD4   | 28.31 | 28.32   | 0.010       | 3.000E+04              | 4.477                 | 3.00E+04 | 0.00E+00    |
| D03  | VIC   |        | Unkn-04 | 9.3_D  | 27.28 | 27.25   | 0.042       | 6.386E+04              | 4.805                 | 6.52E+04 | 1.86E+03    |
| D04  | VIC   |        | Unkn-04 | 9.3_D  | 27.22 | 27.25   | 0.042       | 6.649E+04              | 4.823                 | 6.52E+04 | 1.86E+03    |
| D05  | VIC   |        | Unkn-12 | 10.2_C | 27.79 | 27.76   | 0.040       | 4.526E+04              | 4.656                 | 4.62E+04 | 1.26E+03    |
| D06  | VIC   |        | Unkn-12 | 10.2_C | 27.73 | 27.76   | 0.040       | 4.704E+04              | 4.673                 | 4.62E+04 | 1.26E+03    |
| D11  | VIC   |        | Unkn-36 | 5.6_B  | 27.08 | 27.05   | 0.041       | 7.340E+04              | 4.866                 | 7.49E+04 | 2.06E+03    |
| D12  | VIC   |        | Unkn-36 | 5.6_B  | 27.02 | 27.05   | 0.041       | 7.632E+04              | 4.883                 | 7.49E+04 | 2.06E+03    |
| E01  | VIC   |        | Std-05  | STD5   | 31.85 | 31.80   | 0.073       | 3.000E+03              | 3.477                 | 3.00E+03 | 0.00E+00    |
| E02  | VIC   |        | Std-05  | STD5   | 31.75 | 31.80   | 0.073       | 3.000E+03              | 3.477                 | 3.00E+03 | 0.00E+00    |
| E03  | VIC   |        | Unkn-05 | 12.5_D | 27.28 | 27.23   | 0.070       | 6.402E+04              | 4.806                 | 6.62E+04 | 3.13E+03    |
| E04  | VIC   |        | Unkn-05 | 12.5_D | 27.18 | 27.23   | 0.070       | 6.845E+04              | 4.835                 | 6.62E+04 | 3.13E+03    |
| E05  | VIC   |        | Unkn-13 | 6.5_C  | 27.44 | 27.43   | 0.004       | 5.757E+04              | 4.760                 | 5.77E+04 | 1.44E+02    |
| E06  | VIC   |        | Unkn-13 | 6.5_C  | 27.43 | 27.43   | 0.004       | 5.777E+04              | 4.762                 | 5.77E+04 | 1.44E+02    |
| E07  | VIC   |        | Unkn-21 | 3.4_D  | 27.08 | 27.05   | 0.043       | 7.312E+04              | 4.864                 | 7.47E+04 | 2.19E+03    |
| E08  | VIC   |        | Unkn-21 | 3.4_D  | 27.02 | 27.05   | 0.043       | 7.621E+04              | 4.882                 | 7.47E+04 | 2.19E+03    |
| E09  | VIC   |        | Unkn-29 | 10.5_A | 27.54 | 27.45   | 0.124       | 5.372E+04              | 4.730                 | 5.71E+04 | 4.80E+03    |
| E10  | VIC   |        | Unkn-29 | 10.5_A | 27.36 | 27.45   | 0.124       | 6.052E+04              | 4.782                 | 5.71E+04 | 4.80E+03    |
| E11  | VIC   |        | Unkn-37 | 9.2_D  | 27.68 | 27.62   | 0.085       | 4.889E+04              | 4.689                 | 5.10E+04 | 2.94E+03    |
| E12  | VIC   |        | Unkn-37 | 9.2_D  | 27.56 | 27.62   | 0.085       | 5.305E+04              | 4.725                 | 5.10E+04 | 2.94E+03    |
| F01  | VIC   |        | Std-06  | STD6   | 34.72 | 35.14   | 0.587       | 3.000E+02              | 2.477                 | 3.00E+02 | 0.00E+00    |
| F02  | VIC   |        | Std-06  | STD6   | 35.55 | 35.14   | 0.587       | 3.000E+02              | 2.477                 | 3.00E+02 | 0.00E+00    |
| F05  | VIC   |        | Unkn-14 | 5.4_A  | 27.34 | 27.68   | 0.485       | 6.160E+04              | 4.790                 | 5.01E+04 | 1.62E+04    |
| F06  | VIC   |        | Unkn-14 | 5.4_A  | 28.02 | 27.68   | 0.485       | 3.864E+04              | 4.587                 | 5.01E+04 | 1.62E+04    |

Quantification Data

| Well | Fluor | Target | Content | Sample | Cq    | Cq Mean | Cq Std. Dev | Starting Quantity (SQ) | Log Starting Quantity | SQ Mean  | SQ Std. Dev |
|------|-------|--------|---------|--------|-------|---------|-------------|------------------------|-----------------------|----------|-------------|
| F07  | VIC   |        | Unkn-22 | 13.1_B | 27.04 | 27.05   | 0.008       | 7.526E+04              | 4.877                 | 7.50E+04 | 3.89E+02    |
| F08  | VIC   |        | Unkn-22 | 13.1_B | 27.05 | 27.05   | 0.008       | 7.471E+04              | 4.873                 | 7.50E+04 | 3.89E+02    |
| F09  | VIC   |        | Unkn-30 | 6.2_D  | 26.36 | 26.40   | 0.063       | 1.198E+05              | 5.079                 | 1.16E+05 | 5.00E+03    |
| F10  | VIC   |        | Unkn-30 | 6.2_D  | 26.45 | 26.40   | 0.063       | 1.128E+05              | 5.052                 | 1.16E+05 | 5.00E+03    |
| F11  | VIC   |        | Unkn-38 | 13.2_A | 27.72 | 28.04   | 0.460       | 4.753E+04              | 4.677                 | 3.90E+04 | 1.20E+04    |
| F12  | VIC   |        | Unkn-38 | 13.2_A | 28.37 | 28.04   | 0.460       | 3.052E+04              | 4.485                 | 3.90E+04 | 1.20E+04    |
| G05  | VIC   |        | Unkn-15 | 2.3_B  | 27.66 | 27.60   | 0.086       | 4.924E+04              | 4.692                 | 5.14E+04 | 2.99E+03    |
| G06  | VIC   |        | Unkn-15 | 2.3_B  | 27.54 | 27.60   | 0.086       | 5.346E+04              | 4.728                 | 5.14E+04 | 2.99E+03    |
| G07  | VIC   |        | Unkn-23 | 12.4_C | 26.29 | 26.68   | 0.548       | 1.252E+05              | 5.098                 | 9.96E+04 | 3.63E+04    |
| G08  | VIC   |        | Unkn-23 | 12.4_C | 27.07 | 26.68   | 0.548       | 7.395E+04              | 4.869                 | 9.96E+04 | 3.63E+04    |
| G09  | VIC   |        | Unkn-31 | 5.2_D  | 27.16 | 27.09   | 0.104       | 6.933E+04              | 4.841                 | 7.30E+04 | 5.15E+03    |
| G10  | VIC   |        | Unkn-31 | 5.2_D  | 27.01 | 27.09   | 0.104       | 7.662E+04              | 4.884                 | 7.30E+04 | 5.15E+03    |
| H01  | VIC   |        | NTC-01  | NEG    | N/A   | 0.00    | 0.000       | N/A                    | N/A                   | 0.00E+00 | 0.00E+00    |
| H02  | VIC   |        | NTC-01  | NEG    | 39.66 | 39.66   | 0.000       | N/A                    | N/A                   | N/A      | 0.00E+00    |
| H03  | VIC   |        | Unkn-08 | 9.3_B  | 27.60 | 27.58   | 0.028       | 5.138E+04              | 4.711                 | 5.21E+04 | 1.01E+03    |
| H04  | VIC   |        | Unkn-08 | 9.3_B  | 27.56 | 27.58   | 0.028       | 5.280E+04              | 4.723                 | 5.21E+04 | 1.01E+03    |
| H05  | VIC   |        | Unkn-16 | 7.4_B  | 27.31 | 27.40   | 0.124       | 6.275E+04              | 4.798                 | 5.92E+04 | 4.99E+03    |
| H06  | VIC   |        | Unkn-16 | 7.4_B  | 27.48 | 27.40   | 0.124       | 5.570E+04              | 4.746                 | 5.92E+04 | 4.99E+03    |
| H09  | VIC   |        | Unkn-32 | 6.3_C  | 27.44 | 27.45   | 0.011       | 5.730E+04              | 4.758                 | 5.70E+04 | 4.35E+02    |
| H10  | VIC   |        | Unkn-32 | 6.3_C  | 27.46 | 27.45   | 0.011       | 5.668E+04              | 4.753                 | 5.70E+04 | 4.35E+02    |
| H11  | VIC   |        | Unkn-40 | 15.5_D | 28.36 | 28.39   | 0.042       | 3.073E+04              | 4.488                 | 3.01E+04 | 8.61E+02    |
| H12  | VIC   |        | Unkn-40 | 15.5_D | 28.42 | 28.39   | 0.042       | 2.951E+04              | 4.470                 | 3.01E+04 | 8.61E+02    |

Bar Chart

Expression analysis requires that at least two wells contain sample names, targets, and valid Cqs.

QC Parameters

Data

| Description                             | Value | Use  | Results | Exclude Wells | All excluded wells |
|-----------------------------------------|-------|------|---------|---------------|--------------------|
| Negative control with a Cq less than    | 38    | True |         | False         |                    |
| NTC with a Cq less than                 | 38    | True |         | False         |                    |
| NRT with a Cq less than                 | 38    | True |         | False         |                    |
| Positive control with a Cq greater than | 30    | True |         | False         |                    |
| Unknown without a Cq                    | N/A   | True |         | False         |                    |

Data

| Description                             | Value | Use  | Results                                                                                                | Exclude Wells | All excluded wells |
|-----------------------------------------|-------|------|--------------------------------------------------------------------------------------------------------|---------------|--------------------|
| Standard without a Cq                   | N/A   | True |                                                                                                        | False         |                    |
| Efficiency greater than                 | 110.0 | True |                                                                                                        |               |                    |
| Efficiency less than                    | 90.0  | True |                                                                                                        |               |                    |
| Std Curve R^2 less than                 | 0.980 | True |                                                                                                        |               |                    |
| Replicate group Cq Std Dev greater than | 0.20  | True | FAM:C11, C12, E9, E10, F1, F2, F5, F6, F11, F12, G7, G8. VIC:B3, B4, F1, F2, F5, F6, F11, F12, G7, G8. | False         |                    |